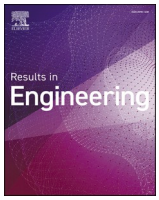
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Advancing water quality assessment and prediction using machine learning models, coupled with explainable artificial intelligence (XAI) techniques like shapley additive explanations (SHAP) for interpreting the black-box nature

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ABSTRACT

Water quality assessment and prediction play crucial roles in ensuring the sustainability and safety of freshwater resources. This study aims to enhance water quality assessment and prediction by integrating advanced machine learning models with XAI techniques. Traditional methods, such as the water quality index, often require extensive data collection and laboratory analysis, making them resource-intensive. The weighted arithmetic water quality index is employed alongside machine learning models, specifically RF, LightGBM, and XGBoost, to predict water quality. The models' performance was evaluated using metrics such as MAE, RMSE, R^2, and R. The results demonstrated high predictive accuracy, with XGBoost showing the best performance (R^2 = 0.992, R = 0.996, MAE = 0.825, and RMSE = 1.381). Additionally, SHAP were used to interpret the model's predictions, revealing that COD and BOD are the most influential factors in determining water quality, while electrical conductivity, chloride, and nitrate had minimal impact. High dissolved oxygen levels were associated with lower water quality index, indicative of excellent water quality, while pH consistently influenced predictions. The findings suggest that the proposed approach offers a reliable and interpretable method for water quality prediction, which can significantly benefit water specialists and decision-makers.

1. Introduction

Water is an essential resource for sustaining life and occupies two-thirds of the Earth's surface. Nevertheless, the amount of freshwater availability remains limited, with a substantial number being surface water globally. However, this surface water is constantly deteriorating due to both point and non-point sources, aggravating the existing situation. This degradation may ultimately result in an impairment in the accessibility of drinking water, which will possess an impact on the economy and require deliberate remedies [1–4]. Rivers, which serve as the primary water source for numerous purposes including industrial and irrigation, are particularly susceptible to environmental

Abbreviations: XAI, Explainable Artificial Intelligence; SHAP, SHapley Additive exPlanations; RF, Random Forest; LightGBM, Light Gradient Boosting Machine; XGBoost, Extreme Gradient Boosting; MAE, Mean Absolute Error; RMSE, Root Mean Square Error; R^2, Coefficient of Determination; R, Pearson Correlation Coefficient; COD, Chemical Oxygen Demand; BOD, Biochemical Oxygen Demand.

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degradation due to their constantly evolving characteristics and interaction with waste disposal. Water quality pertains to the overall state or condition of water, including its chemical, physical, and biological aspects. Hence, it is crucial to accurately predict and control the quality of surface water [5–9].

The assessment of water quality involves the collection and analysis of comprehensive data, including numerous water quality indicators. Summarizing overall water quality using a single parameter remains challenging. To address this, various methods, such as the water quality index (WQI) model, have been developed. Initially established by Horton in 1960, the WQI is based on 10 key water quality parameters deemed crucial for assessing the condition of a water body [10]. The WQI calculates a single value that reflects the usability of water by assessing temporally and spatially varying water quality parameters. Over time, several WQI models have been developed worldwide, including the United States National Sanitation Foundation Water Quality Index (NSFWQI) [11], the Scottish Research Development Department Water Quality Index (SRDDWQI) [11], the Canadian Council of Ministers of the Environment Water Quality Index (CCMEWQI) [11,12], Weighted Arithmetic Water Quality Index (WAWQI) [13,14] and the British Columbia Water Quality Index (BCWQI) [11]., among others.

However, manual computations and laboratory analyses of extensive water quality data can be labor-intensive, inefficient, and costly. This challenge has led to the growing use of intelligent techniques, such as machine learning (ML), particularly in scenarios requiring real-time predictions [15–17]. ML enables systems to learn from experience and improve their performance without complex programming [18]. As a crucial component of artificial intelligence (AI), ML has seen increasing application in various scientific fields, including water quality assessment [19]. Recent studies, such as those by Ghosh, H. et al. (2023), have employed multiple ML models to predict water quality [20]. Additionally, Haghiabi, A. H. et al. (2018) explored the effectiveness of various ML models, including Artificial Neural Networks (ANN), Group Method of Data Handling (GMDH), and Support Vector Machine (SVM), in forecasting water quality [21]. Similarly, research by Nasir, N. et al. (2022), Bui, D. T. et al. (2020), and Asadollah, S. B. H. S. et al. (2021) utilized different ML models to evaluate water quality in terms of WQI [22–24].

One major drawback of sophisticated ML models is their intrinsic lack of interpretability, often referred to as their “black-box” nature. This leads to it challenging to understand and interpret the outcomes and underlying reasoning acquired from studies [25,26]. Although ML models are capable of learning intricate functions and relationships, they lack the ability to provide insights into their internal workings, making it challenging for researchers and end users who are interested in understanding the process and reasoning behind the predictions. The absence of transparency and interpretability has a profound effect on numerous sensitive areas, such as water quality, which is vital for the survival of all living creatures. Because of this rationale, both the authorities and the end users frequently prefer to depend on conventional approaches that offer transparency and comprehensibility, ignoring the superior performance of ML models. Hence, it is crucial to incorporate transparency and the explainability into complicated ML models. This allows not only for accurate predictions, but also for domain experts and end users to understand and believe these algorithms. As a result, they can comprehend the underlying factors that contribute to the model’s outcomes.

XAI emerged with the aim of addressing the challenge of interpretability that arises from the black-box nature of sophisticated ML models [25,26]. XAI enables researchers to acquire insights that contribute to a more profound comprehension of different aspects, such as the importance of features, their interdependencies, and other pertinent aspects which valuable for decision making process. Importantly, it enables the establishment of an understandable and reasonable justification that can be effectively communicated to the end user [26].

Hence, this comprehensive study attempts to achieve the following objectives: (1) to assess and predict water quality in relation to the WQI using three machine learning models: RF, XGBoost, and LightGBM; (2) to employ the SHAP method to interpret the black-box nature of these machine learning models and understand the impact of each water quality parameter on the corresponding predictions [27–33]; and (3) to enhance the interpretability and transparency of machine learning models through the incorporation of XAI techniques, thereby strengthening the reliability of the results and supporting informed decision-making in water quality evaluation and management. Through these objectives, the study endeavors to provide a robust, interpretable, and transparent framework for water quality prediction that can be trusted by both researchers and decision-makers.

2. Materials and methodology

The approach utilized in this investigation is illustrated in Fig. 1. Starting from section 2.2, the authors have presented an extensive description of the workflow.

2.1. Dataset and processing

Water quality data was collected from twelve sampling points near to the Thalduwa, Hanwella, Kaduwela, Welivita, Victoria, Seethawaka, Pugoda (Ferry and Ela), Wak Oya, Maha Oya, Pusseli Oya and Raggahawatte as illustrated in Fig. 2. The data was acquired from the Central Environmental Authority (CEA) of Sri Lanka and is readily accessible on their website. The dataset encompasses eight water quality parameters, namely pH, electrical conductivity (EC), dissolved oxygen (DO), chemical oxygen demand (COD), biochemical oxygen demand (BOD), chloride (Cl^-), total nitrate (NO_3^-), and total phosphate (PO_4^{2-}).

In advance of doing the analysis, the data undergone preprocessing, which involved the initial phase of outlier detection. The Z-score method was employed to exclude outliers from the data. This method quantifies the extent to which the data deviates from the mean in terms of standard deviation. In addition, the water quality characteristics were graphically represented in order to analyze the distribution of the data and detect any potential outliers, as illustrated in Fig. 3. Following the identification and elimination of outliers, the dataset undertook removal of missing values. This was done by the authors with the intention of preserving the originality of the data and because of the proportion of missing values was less than 10 % of the dataset, no analysis was conducted to impute these missing values. Annexure A1 presents the fundamental statistical parameters obtained after doing the preprocessing of the dataset.

2.2. WQI model development

The WAWQI was employed to compute the WQI utilizing the above specified water quality parameters. Several studies [13,14,34–36] made use of this method to categorize water quality according to its level of purity, using widely identified indicators of water quality. The calculation of WAWQI can be determined using the equations depicted in Annexure B [37]. The water quality status is assessed by considering the WQI values and presented in Table 1 [37].

The water quality standards for drinking water, including SLS 614, WHO, and BIS, are outlined in Annexure C1.

2.3. Development of ML models

The authors initially examined the correlation matrix (Fig. 4) to determine the relationships between the dataset, which consists of water quality parameters, and the target variable, which is the WAWQI. The correlation matrix and pairwise correlation depicted in Annexure F suggest that in many cases, there is no evident correlation among the variables in the dataset, as well as between the features and the target

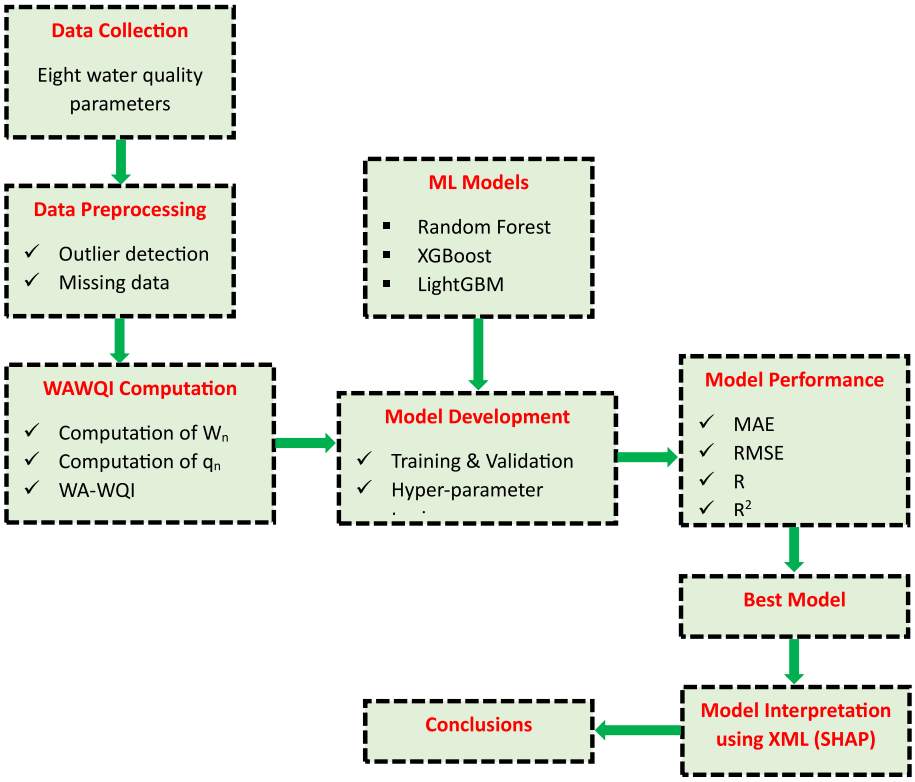


Fig. 1. Methodology of the study.

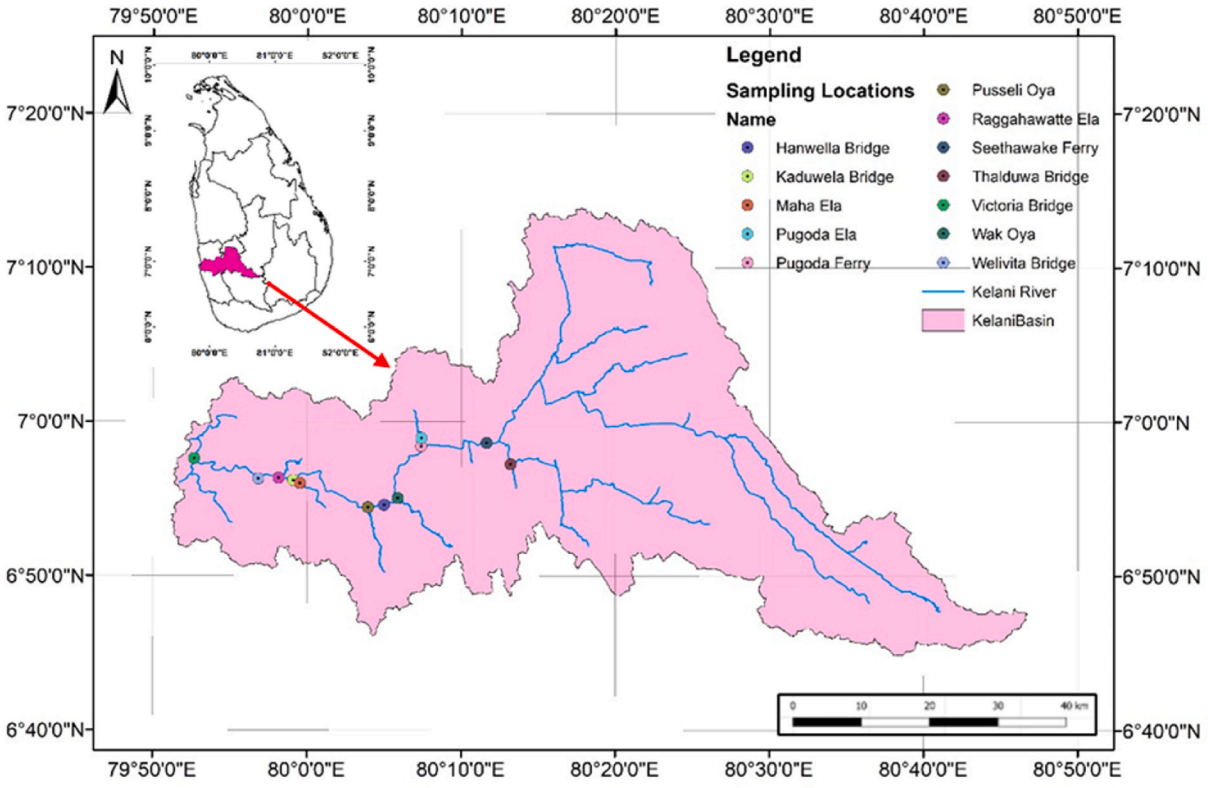


Fig. 2. Sampling points located across the Kelani River basin.

variable. Therefore, it may be concluded that these parameters are independent of each other. However, there was a strong positive correlation (R = 0.83) between two water quality measures, particularly Cl^- and EC. Furthermore, there was a positive correlation between the WQI and BOD with a correlation coefficient (R) of 0.64, and similarly between WQI and COD (R = 0.81). On the other hand, there was a

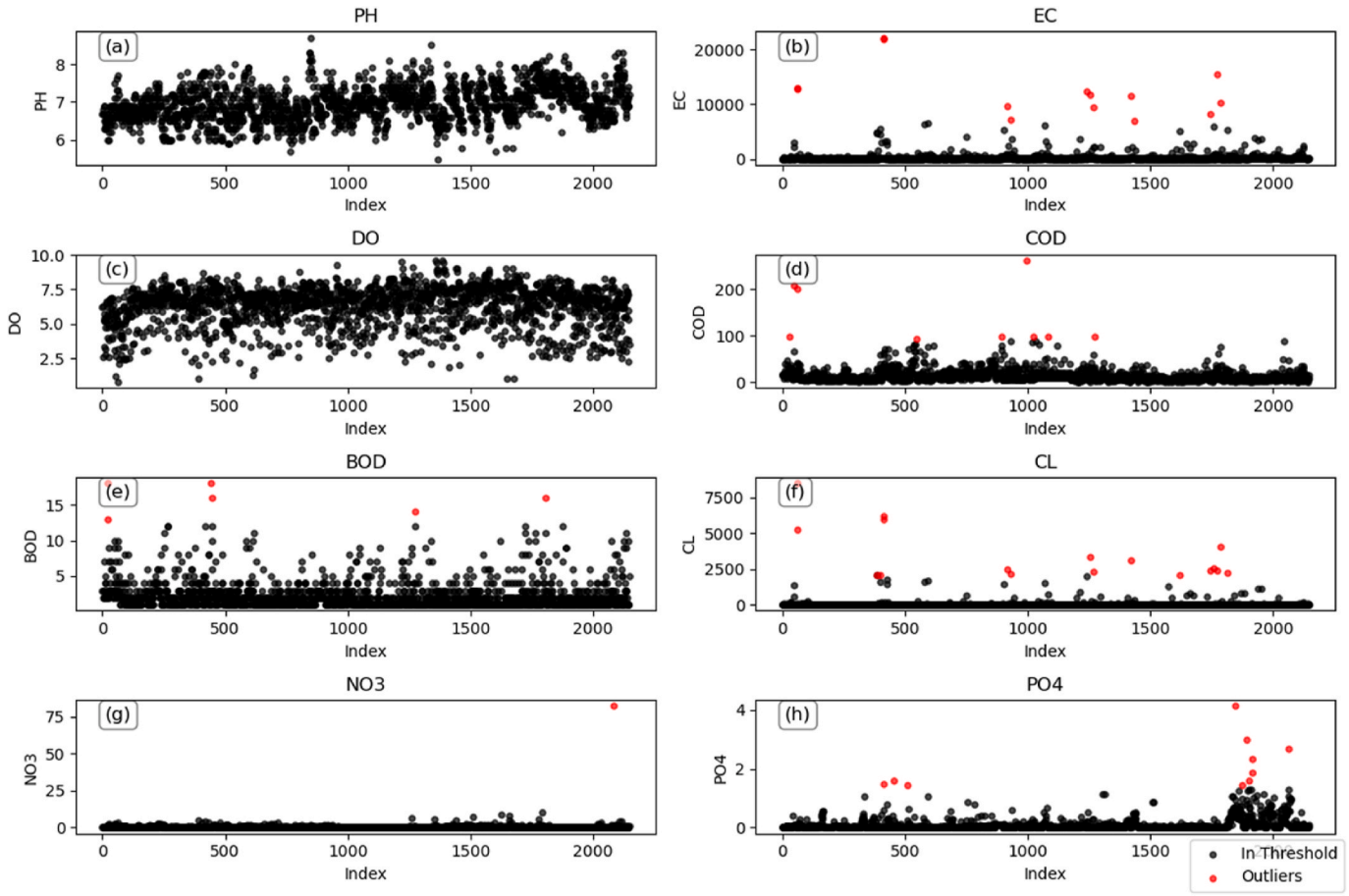


Fig. 3. Outliers (in red dots) detected in each water quality parameter.

Table 1
Recommended range of WQI for intended usage.

WQI	Water quality status	Intended usage		
		Drinking	Irrigation	Industrial
0–25	Excellent	✓	✓	✓
26–50	Good	✓	✓	✓
51–75	Poor	×	✓	✓
76–100	Very poor	×	✓	×
>100	Not suitable for drinking	Proper treatment is required		

moderate negative correlation (R = −0.51) between WQI and DO. Still, the remaining characteristics do not have an adequate correlation, therefore it was not feasible to establish a linear regression model for predicting water quality. Nevertheless, the WQI may exhibit a non-linear correlation with the water quality parameters. In this study, the authors choose to investigate this relationship by employing sophisticated predictive models, such as ML models.

ML models are algorithms designed to evaluate data and discern intricate patterns and relationships in order to make predictions about future events. This study applied three tree-based machine learning models, RF, XGBoost, and LightGBM, due to their simplicity and powerful conceptual frameworks that can effectively handle complicated nonlinear relationships between the target and features [38]. Each of these models has different levels of complexity and explainability, which means that post-hoc explanations are needed to clarify how they work and to get insights into the components that affect their predictions.

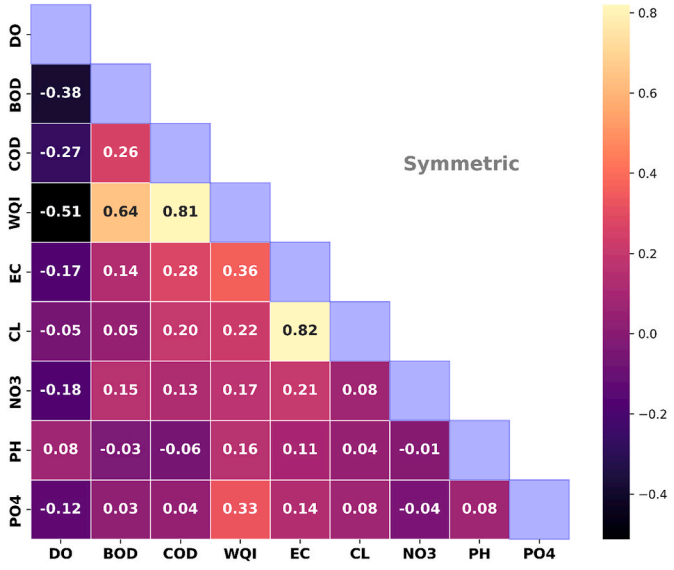


Fig. 4. Correlation matrix of the dataset.

2.3.1. *RF regressor*

RF is a method of ensemble learning introduced by Breiman (2001) [39]. RF has demonstrated excellent performance with minimal computer resources. The fundamental component of RF is known as base learners, which are usually developed using the classification and regression tree (CART) methodology [40]. RF is often composed of an

ensemble of numerous distinct trees, with each tree being grown using a bootstrap sampling of the original data. One key distinction between RF trees and CART is that RF trees are created in a non-deterministic manner utilizing a two-stage randomization approach. The original data is first sampled using bootstrap sampling, and then randomization is applied at the node level of the tree in RF. RF selects a random subset of variables, and only those variables are considered as candidates to determine the optimum split for each node. The objective of this two-step randomization process is to eliminate any correlation between the trees, resulting in a forest ensemble with minimal variance, which is referred to as a bagging phenomenon.

In general, random forest regression involves randomly drawing several samples from a given dataset and constructing CART models. Ultimately, the outcomes of all these trees are merged, and an approximation of the target variables is obtained by calculating the average of the various tree outputs. Thus, the randomization process employed in random forest regression ensures robustness against noise and outliers in the dataset. Since there is no relationship between individual trees and each tree is allowed to grow to its maximum size, the likelihood of overfitting is reduced compared to a single tree. This study utilized a specific set of hyper-parameters that were precisely calibrated in order to train the model using GridSearchCV, as demonstrated in Annexure D1.

2.3.2. XGBoost regressor

XGBoost, is an integration algorithm that combines gradient boosting techniques [41]. It was proposed by Chen and Guestrin in 2016 [42] as a method to predict a response variable based on particular variables. The algorithm employs a recursive binary splitting technique to determine the optimal split at each iteration, ultimately leading to the most optimal model. The fundamental concept behind this approach is that it constructs a series of CARTs sequentially, with each consecutive tree model being trained using the residuals of the previous tree. In a nutshell, the new model rectifies the mistakes caused by the previously trained tree and subsequently forecasts the result. The tree-based structure of XGBoost is not affected by outliers and, similar to other boosting techniques, is resistant to overfitting, which greatly simplifies the process of selecting the model. Equation (1) provides an expression for the objective function, which takes into account both the loss and regularization terms.

$$\textit{Objective Function} = \sum_{j=1}^{N} l\left(y_j, \hat{y}_j\right) + \sum_{j=1}^{K} \Omega\left(f_j\right) \tag{1}$$

The expression $l\left(y_j, \hat{y}_j\right)$ denotes the loss function, which quantifies the disparity between the observed response y_j and the predicted response $\hat{y}_j$. The regularization term $\Omega\left(f_j\right) = \gamma T + \frac{1}{2}\lambda\|\omega\|^2$ commonly referred to as the shrinkage penalty because it penalises the complexity of the model and contributes to avoiding overfitting. The parameter γ in tree pruning controls the depth of the tree, while T is the number of nodes in the hidden layer of each neural network. The parameter λ is the regularization parameter related with the l2-norm of the scores vector (ω)., which measures the magnitude of scores. Pruning able to mitigate the intricacy of the model and, consequently, rectifies overfitting. In Annexure D1, the authors have provided a detailed explanation of the hyper-parameters utilized in this study, along with the corresponding optimized values.

2.3.3. LightGBM regressor

LightGBM is a tree-based algorithm integrated with boosting [43]. The LightGBM algorithm differs from the XGBoost algorithm by incorporating two innovative data sampling and classification methods, namely Exclusive Feature Bundling (EFB) and Gradient-based One-Side Sampling (GOSS) [43]. Additionally, LightGBM replaces the inefficient level-wise growth strategy commonly used in decision trees with a more efficient leaf-wise growth adaptation. Due to its distinctive combination of characteristics, this method efficiently and precisely carries out data scanning, sampling, clustering, and classification in a shorter timeframe compared to other boosting algorithms including XGBoost. When considering factors such as memory usage, processing time, and arithmetic speed, LightGBM is an ideal choice for achieving faster training, efficient performance, optimal memory utilization, decent precision, parallelism, and the ability to process large-scale data. The hyper-parameters utilized to train the LightGBM model and their optimized values obtained from GridSearchCV are listed in Annexure D1.

2.4. Evaluation of ML models performance

ML model's performance was evaluated using four statistical indices including MAE, RMSE, R^2, and R. These metrics are calculated utilizing equations illustrated in Annexure E.

Furthermore, visual comparisons were used for the purpose of model evaluation. Scatter plots are often employed techniques for analyzing the distribution of datasets and evaluating model performance [44,45] as it can be used analyze the density and distribution of data structures. In this work, the authors employed ±10% and ±20% deviation lines to evaluate the disparity between the predicted values and the measured WQI values.

2.5. ML models interpretation using XAI

XAI lacks a general definition and might be described as an explanatory agent that elucidates the underlying reasoning behind model predictions [46]. XAI can be categorized based on model interpretability, applicability and scope. Model interpretability can be classified depending on whether the inner workings of the model are easily understandable (i.e. model-based) or if a specific explainable technique is needed to interpret the model after its development (i.e. post-hoc). In terms of applicability, there are some XAI approaches that can be used for all sorts of ML models, regardless of their specific characteristics (i.e., model-agnostic). However, there are also techniques that are specific to certain types of ML models and can only be used to those specific models (i.e. model-specific). Lastly, XAI can be segmented based on its scope, which can either be focused on providing an explanation for the entire ML model (i.e. global explanation) or solely focused on explaining a particular component or individual predictions made by the model (i.e. local explanation).

SHAP [47] and Local Interpretable model-agnostic explanation (LIME) [48] are popular XAI methods extensively employed in the field of ML. LIME is widely used for providing local explanations, but SHAP is particularly helpful as it offers both local and global explanations by combining individual local explanations and comparing each instance's prediction with the average prediction. LIME operates on the principle that a model is perceived as intricate on a global scale, but as a linear function on a local scale. LIME generates marginally altered samples within the closest proximity of the data point, which are then inputted into the black-box ML model to produce predictions. Subsequently, LIME employs straightforward and comprehensible models like linear regression, LASSO, or decision tree to calculate interpretable parameters for explanation. SHAP calculates Sharp values to estimate the weights corresponding to the sampled examples. LIME generates dummy values by taking into account the surrounding instances, without including the real feature value of the instance [49]. Conversely, the SHAP value is widely acknowledged as a comprehensive metric for determining the significance of features. Based on the characteristics mentioned above, the authors opted to use the SHAP explainable model in this work to understand the predictions made by the ML models.

2.5.1. SHAP

The SHAP method, introduced by Lundberg and Lee [47], is employed to understand the model's output by computing the Shapley

values using the principles of game theory [50]. Shapley values are employed in ML to measure the individual impact of each feature in the model, which together contribute to the prediction [51]. Lundberg and Lee have developed many versions of SHAP including DeepSHAP, Kernel SHAP, LinearSHAP, and TreeSHAP to approximate SHAP values for different types of ML models. Nevertheless, the authors of this study have employed the TreeSHAP estimation methodology, which is specifically designed for tree-based approaches like random forest and gradient boosting [52]. TreeSHAP offers the benefit of being considerably more efficient, and it additionally enables the assessment of interaction effects. Utilizing the formula $x = h_x(z)$, SHAP generates simplified inputs (z) by mapping feature values (x) to z. Subsequently, a linear function of the binary variables can be employed to approximate the original model $f(x)$ based on z (Equation (2)) [47].

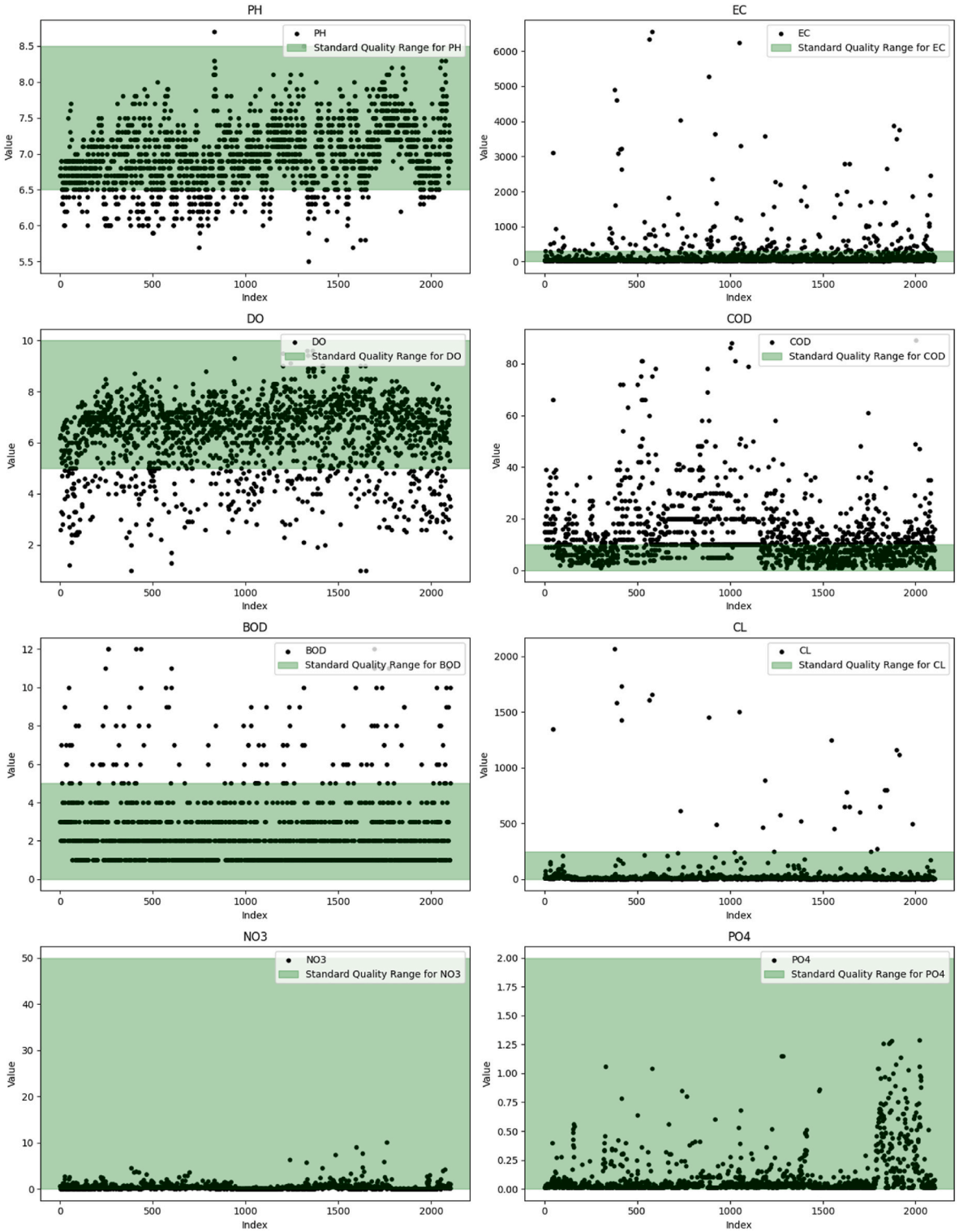


Fig. 5. Distribution of recorded values for water quality parameters and their standard ranges.

$$f(x) = g(z) = \varnothing_o + \sum_{i=1}^{M} \varnothing_i z_i \quad (2)$$

Where, z = {0, 1}^{M}, M is the number of input features, $\varnothing_o = f(h_x(0))$, and $\varnothing_i$ is the feature attribution value.

$$\varnothing_i = \sum_{S \subseteq F\{i\}} \frac{|S|!(M - |S|! - 1)!}{M!} [f_x(S \bigcup \{i\}) - f_x(S)] \quad (3)$$

$$f_x(S) = f\left(h_x^{-1}(z)\right) = E[f(x)|x_S \quad (4)$$

Where, S is the subset of F that does not contain the ith feature, F is the set of non-zero inputs in z, and $\varnothing_i$ is the SHAP value, which is the unified measure of additive feature attributions. The expression $E[f(x)|x_S$ expresses the anticipated value of the function on the subset S.

3. Results and discussion

3.1. Statistical analysis WAWQI and water quality parameters

Water quality parameters used in this study and their distribution along with the standards water quality standards for the human drinking water consumption are illustrated in Fig. 5. It is evident that most of the time the parameters are exceeded standard ranges except nitrate and phosphates. Annexure A1 presents the statistical summary including mean, median, maximum, minimum and standard deviation of water quality parameters. The mean value of parameter such pH (6.96 ± 0.45), EC (167.65 ± 499.33 μS/cm), DO (6.34 ± 1.39 mg/l), COD (13.08 ± 11.67 mg/l), BOD (2.18 ± 1.80 mg/l), Cl^- (28.49 ± 136.83 mg/l), NO_3^- (0.41 ± 0.69 mg/l), and PO_4^{3-} (0.09 ± 0.18 mg/l). Makubura et al., 2022 has provided a comprehensive analysis about the water quality in the Kelani River basin utilizing above water quality parameters [13].

The calculated WAWQI values ranges between 11.57 and 137.08 suggesting that the water quality of the Kelani River basin covers all the water quality status, indicates that the water quality of the Kelani River basin encompasses a wide range of changes, from excellent to not acceptable for drinking, as depicted in Fig. 6. The elevated WQI can be attributed to unfavorable levels of water quality indicators, mainly due to the inflow of industrial wastewater from the Biyagama industrial zone and Seethawaka industrial zone [13]. Once the WAWQI has been calculated, the data has been utilized as the target variable for the machine learning models to forecast the WQI.

3.2. Performance evaluation of ML models

The ML models were assessed using four statistical parameters: MAE, RMSE, R^2, and R, as illustrated in Table 2. Following a thorough analysis of the findings, it was evident that XGBoost exhibited superior performance compared to the other ML models during both the training and validation stages. When considering the performance based on MAE during the training phase, XGBoost performed the best with a MAE of 0.203, followed by LightGBM with a MAE of 0.743, and RF with a MAE of 0.809. A similar pattern was observed during the validation phase, where XGBoost had a MAE of 0.825, LightGBM had a MAE of 0.831, and RF had a MAE of 1.506.

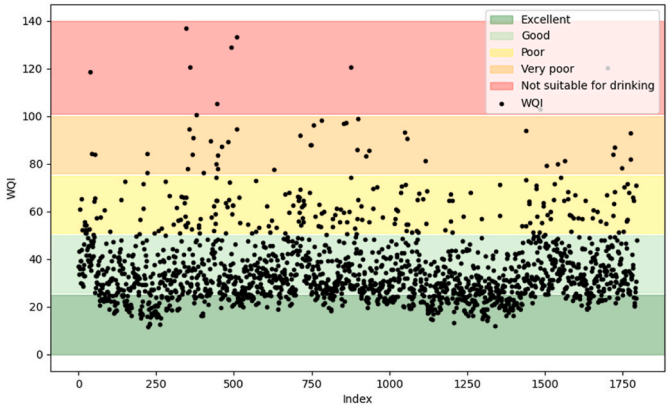


Fig. 6. Computed WAWQI values, together with the corresponding water quality status.

In terms of RMSE, XGBoost again demonstrated superior performance with RMSE values of 0.259 during the training phase and 1.381 during the validation phase. LightGBM followed with RMSE values of 1.375 in training and 1.436 in validation, while RF had RMSE values of 1.486 in training and 2.427 in validation. Moreover, the XGBoost model showed excellent performance with an R^2 value of 0.999 in the training stage and 0.992 in the validation stage. LightGBM and XGBoost exhibited similar performance based on R, with R values of approximately 0.999 and 0.996, respectively, in the training stage, and both achieving an R value of 0.996 in the validation phase.

Therefore, as a conclusion to the performance analysis using these statistical indices, XGBoost consistently outperformed both LightGBM and RF across all measures, making it the most effective model for forecasting the WQI. This finding aligns with the study by Nasir et al. (2022) [22], where another boosting algorithm, CATBoost, was found to be the most accurate in classifying water quality with a high precision rate.

In addition to the performance evaluation based on statistical parameters Fig. 7 depicts the scatter plots along with the line plot to visually interpret the obtained and predicted WQI values for both training (left side) and validation (right side) stages. Based on Fig. 7, it is noticeable that XGBoost model is the best suited ML model for evaluating the WQI as majority of the data points scattered around the best fit line. Still, both LightGBM and RF models achieved excellent accuracy in the training and validation stages, as seen by the WQI values closely aligning with the best fit line and falling within a ±20 % deviation range.

3.3. Interpretation of model predictions using SHAP

3.3.1. SHAP global explanation

The SHAP explanation was employed to analyze the underlying process of the outcomes provided by the XGBoost model, which exhibited superior performance in comparison to other ML models. Fig. 8a depicts the mean absolute SHAP values (i.e. feature importance) and SHAP summary plot (Fig. 8b) of the XGBoost model. Based on the feature importance in relation to the overall model predictions, COD exhibited the highest mean SHAP value of 6.92, indicating that it is the most important contributing element in predicting WQI. BOD ranked second in the list, with a mean SHAP value of 4.45, whereas pH (2.29), PO_4^{3-} (2.16) and DO (2.01) had a nearly equal impact on the model predictions. The effect of EC (0.28), Cl^- (0.16) and NO_3^- (0.1) on the overall output was quite minimal compared to other water quality parameters.

As presented in Fig. 8b, the SHAP summary plot illustrates the

Table 2
Comparison of ML model's performance based on Statistical metrics.

ML Model	Phase	Statistical Metrics			
		MAE	RMSE	R^2	R
RF	Training	0.809	1.486	0.991	0.996
	Validation	1.506	2.427	0.974	0.988
XGBoost	Training	0.203	0.259	0.999	0.999
	Validation	0.825	1.381	0.992	0.996
LightGBM	Training	0.743	1.375	0.992	0.996
	Validation	0.831	1.436	0.991	0.996

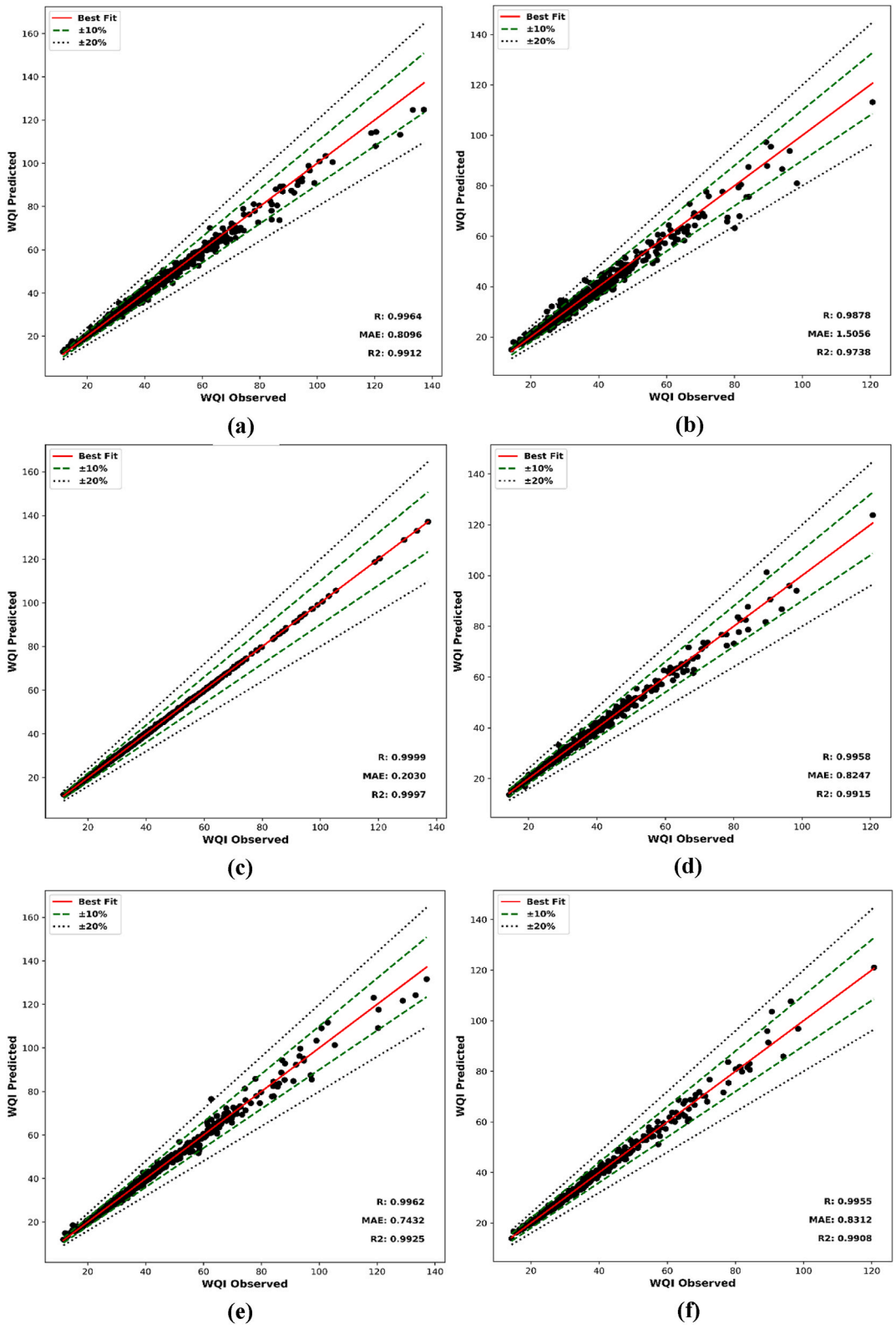


Fig. 7. Scatter plot between WQI observed and WQI predicted using different ML models for training (left) and validation data (right). (a), (b): RF, (c), (b): XGBoost and (e), (f): LightGBM.

relationship between the contribution of each feature value and its magnitude. XGBoost determined that the parameter COD has the greatest impact on the WQI, and lower values of COD led to a decrease in WQI values and vice versa. In a nutshell, a low COD leads to a low WQI, indicating excellent drinking water quality. Similar to the COD, BOD, and PO_4^{3-} had influenced positively to the target. When BOD, and PO_4^{3-} are minimal, WQI decreases and vice versa. Conversely, high DO values corresponded to WQI values, and vice versa. This indicates that low DO concentrations unsuitable for drinking water, as they lead to high WQI values. The symmetrical pattern of the pH values suggests that the

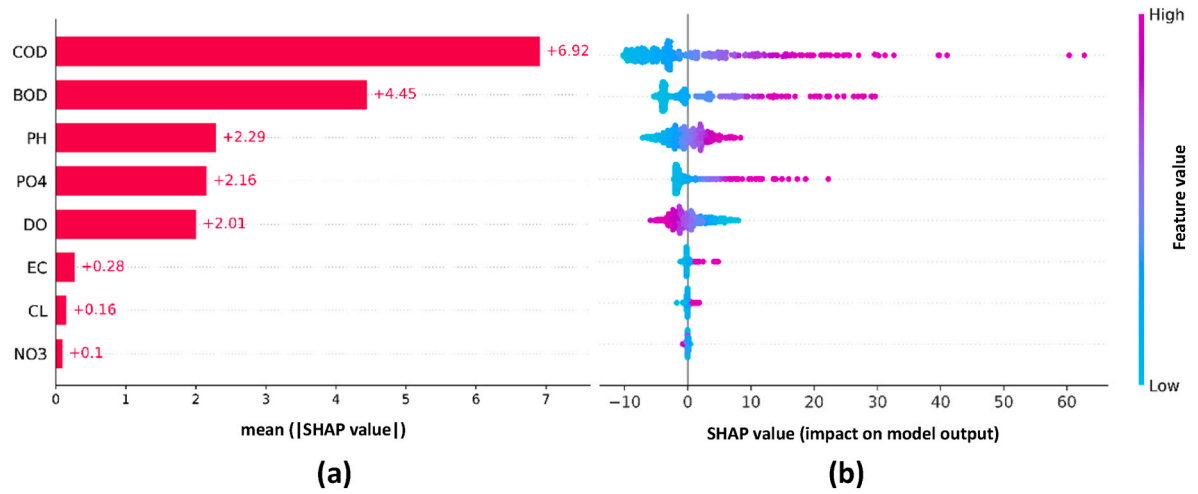


Fig. 8. (a) SHAP feature importance based on mean absolute SHAP values and (b) SHAP summary plot for predictions based on XGBoost.

impact of pH on the WQI remains consistently stable, regardless of the pH level. Interestingly, SHAP effectively elucidates the precise correlations between water quality parameters and overall water quality while referencing to the standard scientific principles in the field. For instance, Sánchez et al. (2007) [53] examined a strong linear relationship between the WQI and the dissolved oxygen deficit, and SHAP has effectively identified and accurately depicted this correlation.

3.3.2. SHAP local explanation

Following the global interpretation, Fig. 9 illustrates the selected scenarios obtained from the SHAP force plot for locally interpreting a specific forecast. The prediction for each plot is shown in bold and is generated by summing all the SHAP values with the base value. Thus, it elucidates that variables make distinct contributions and interactions for each prediction in each given circumstance. Based on Fig. 9a, this scenario has lower COD (10) value pushes the model to a minimal WQI value (25.68). Conversely, higher COD values (21 and 25) correlate with increased WQI (44.02 and 76.70), as seen in Fig. 9b and c. It is important to mention that SHAP has identified DO with a value of 7.9 as causing a decrease in the WQI in Fig. 9a, whereas in Fig. 9c, a DO value of 4 led to an increase in WQI. Similarly, by referring to the force charts, the end-user can gain a comprehensive understanding of the underlying mechanism of each scenario that leads to the outcomes, as depicted in Fig. 9. Furthermore, the authors have included waterfall plots in Annexure G to visually represent the SHAP values for each scenario, allowing end users to thoroughly examine the findings.

The heat map shown in Fig. 10 illustrated the SHAP values distribution of same prediction across initial instances. By default, instances are ordered using a hierarchical clustering algorithm, which groups samples with similar explanations together. However, to simplify this for end-users, the authors have sorted the samples based on descending order of predictions (f(x)), as shown in Fig. 10. Accordingly, the lower features including EC, Cl^- and NO_3^- had minimal impact on the model output, consistent with the global explanation. When f(x) is higher, COD and BOD had the highest positive SHAP values, and vice versa. pH remains constant as there is no significant difference in the SHAP values. Therefore, such strong variations and relationships can be observed using these plots, as described.

4. Conclusions

The primary objective of this study was to assess and predict the water quality in respect to the WAWQI utilization of three ML models. The study also utilized XAI, specifically the SHAP method, to interpret

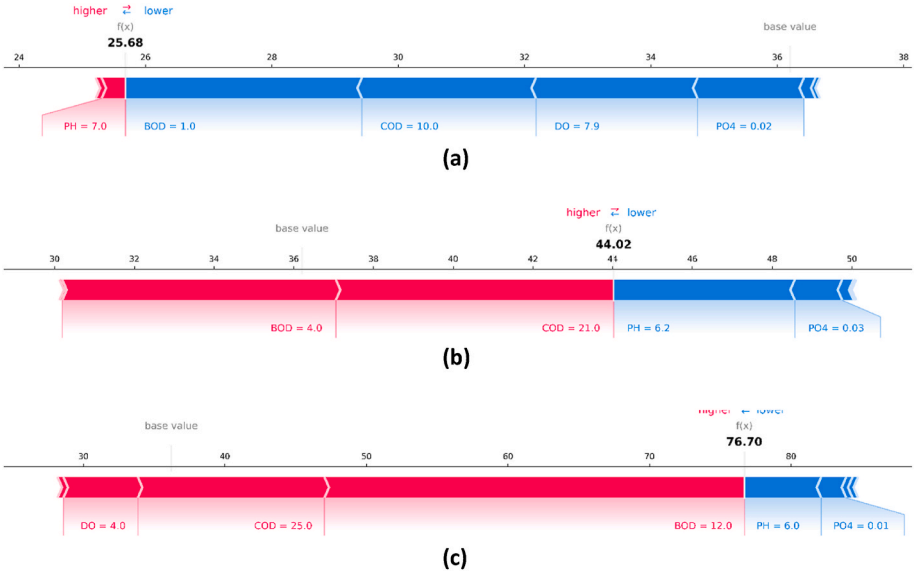


Fig. 9. SHAP force plots for selected scenarios for local interpretation.

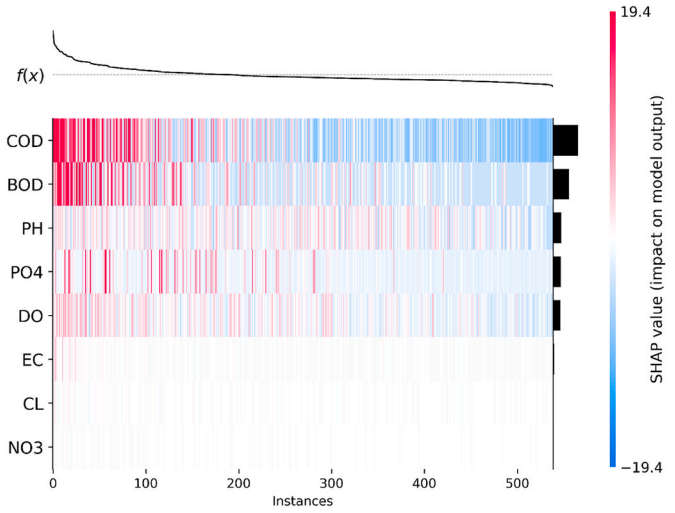


Fig. 10. SHAP heat map plot to visualize the SHAP values distribution of same prediction.

the results. Initially, the WAWQI was determined based on eight water quality parameters: pH, EC, DO, COD, BOD, Cl^-, NO_3^-, and PO_4^{3-}. Subsequently, ML models such as RF, XGBoost, and LightGBM were developed using water quality parameters as features, WAWQI serving as the target variable. The models' performance was later assessed using four statistical indices, namely MAE, RMSE, R^2, and R, as well as scatter plots. Finally, the outcomes obtained from the ML models were explored using SHAP in order to get insight into the underlying reasoning behind the predictions.

The research found that XGBoost performed better, while RF and LightGBM performed well, suggesting the applicability of ML in the field of water quality. The models shown excellent performance in evaluating metrics. The values for R ranged between 0.988 and 0.999, R^2 ranged from 0.974 to 0.992, RMSE ranged from 1.381 to 2.427, and MAE ranged between 0.825 and 1.506. The scatter plots comparing observed and predicted WQI precisely indicates that the predictions did not surpass a 20 % error (i.e. deviation) from the best fit line. The SHAP global explanation analyzed the XGBoost model and determined that COD and BOD have the greatest influence on the WQI, whereas EC, Cl^- and NO_3^- have minor impact on the model's output. The presence of high DO levels led to low WQI values, indicating excellent water quality, and pH had a consistent impact on the predictions, as identified by XGBoost. The SHAP local explanation analyzed three specific cases and clarifies the individual contributions and interactions of the water quality parameters for each prediction in each given scenario. Thus, this study not only evaluated and forecasted the water quality, but the authors also utilized SHAP to analyze the findings. This analysis can be advantageous for end-users as it provides valuable insights that enhance their understanding of various aspects, including the importance of features, their interrelationships, and other relevant factors. These insights are valuable for the decision-making process, as they offer solid justifications and reasoning.

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CRediT authorship contribution statement

Randika K. Makumbura: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Lakindu Mampitiya:** Methodology, Formal analysis. **Namal Rathnayake:** Software, Methodology. **D.P.P. Meddage:** Software, Methodology. **Shagufta Henna:** Writing – review & editing. **Tuan Linh Dang:** Writing – review & editing. **Yukinobu Hoshino:** Writing – review & editing. **Upaka Rathnayake:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. The authors declare no conflict of interest

Table A1
Basic statistical analysis for input parameters

	Unit	Mean	Median	Max	Min	Std
pH	–	6.96	6.90	8.70	5.50	0.45
EC	μS/cm	167.65	40.00	6560.00	10.00	499.33
DO	mg/l	6.34	6.70	9.60	1.00	1.39
COD	mg/l	13.08	10.00	89.00	1.00	11.67
BOD	mg/l	2.18	2.00	12.00	1.00	1.80
Cl^-	mg/l	28.49	7.00	2065.00	0.01	136.83
NO_3^-	mg/l	0.41	0.26	10.06	0.01	0.69
PO_4^{3-}	mg/l	0.09	0.03	1.29	0.01	0.18

Appendix B. Weighted Arithmetic Water Quality Index (WAWQI) Equations

$$WQI = \frac{\Sigma Q_i W_i}{\Sigma W_i} \tag{5}$$

$$Q_i = \frac{100(V_i - V_o)}{(S_i - V_o)} \tag{6}$$

$$W_i = \frac{K}{S_i} \tag{7}$$

$$K = \frac{1}{\sum\left(\frac{1}{S_i}\right)} \tag{8}$$

The variable Q_i represents the quality rating of the ith water quality parameter, while W_i (with the constraint $\Sigma\, W_i = 1$) represents the unit weight of the ith water quality parameter. The variable V_i represents the current value of the ith parameter. The variable V_o represents the desired value of the parameter, which is 0 for all parameters except pH ($V_o = 7$) and DO ($V_o = 14.6$ mg/l). The variable S_i represents the acceptable standard value for the ith parameter. The expression K is a constant that is directly proportional and is determined using Equation (4).

Appendix C

Table C1
Drinking water quality standards based on SLS 614, BIS and WHO

Parameter	Unit	Standards
pH	–	6.5–8.5
DO	mg/l	5
Electrical conductivity	μS/cm	300
Chloride	mg/l	250
Phosphate	mg/l	2
COD	mg/l	10
Nitrate	mg/l	50
BOD	mg/l	5

Appendix D

Table D1
Optimized and assigned hyper-parameters and their values for selected ML models

Random Forest		XGBoost		LightGBM	
Hyper-parameter	Optimized/Assigned	Hyper-parameter	Optimized/Assigned	Hyper-parameter	Optimized/Assigned
n_estimators	**100**	n_estimators	**300**	n_estimators	**500**
criterion	**squared_error**	max_depth	**5**	learning_rate	**0.1**
max_depth	**20**	subsample	**0.5**	boosting_type	**gbdt**
min_samples_split	**2**	gamma	**0.2**	max_depth	**1**
min_samples_leaf	**2**	colsample_bytree	**1**	num_leaves	**31**
n_jobs	**−1**	learning_rate	**0.1**	lambda_l1	**0.5**
random_state	**0**	reg_alpha	**0**		

Appendix E. Equations of Statistical Indices

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|WQI_{mea} - WQI_{pre}\right| \tag{9}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(WQI_{mea} - WQI_{pre}\right)^2} \tag{10}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(WQI_{mea} - WQI_{pre}\right)^2}{\sum_{i=1}^{n}\left(WQI_{mea} - \overline{WQI_{mea}}\right)^2} \tag{11}$$

$$R = \frac{n\sum_{i=1}^{n}\left(WQI_{mea}.WQI_{pre}\right) - \sum_{i=1}^{n}WQI_{mea}\sum_{i=1}^{n}WQI_{pre}}{\sqrt{n\sum_{i=1}^{n}WQI_{mea}{}^{2} - \left(\sum_{i=1}^{n}WQI_{mea}\right)^{2}.n\sum_{i=1}^{n}WQI_{pre}{}^{2} - \left(\sum_{i=1}^{n}WQI_{pre}\right)^{2}}} \tag{12}$$

where WQI_{mea}, WQI_{pre}, and $\overline{WQI_{mea}}$ are WQI measured, WQI predicted and mean values of measured WQI, respectively.

Appendix F



Fig. F1. Pairwise correlation between the features and the target variable.

Appendix G

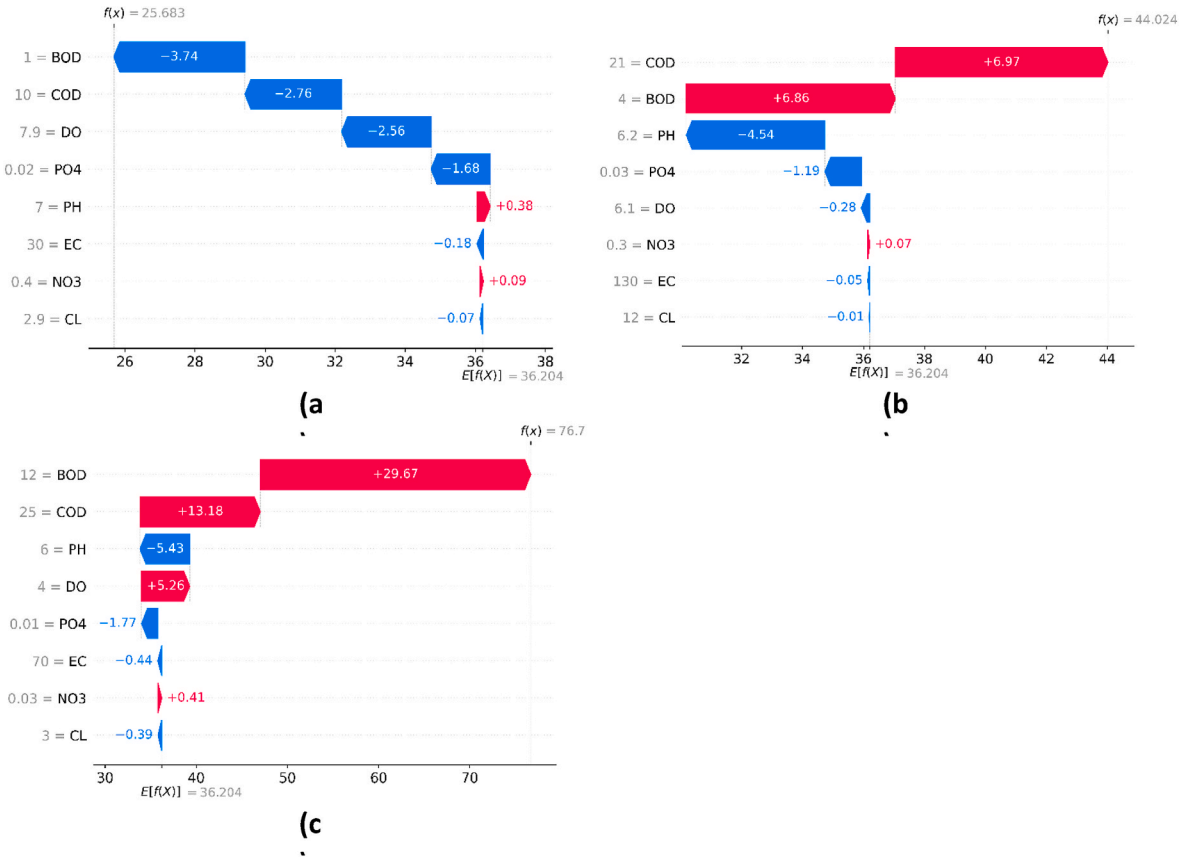


Fig. G1. SHAP waterfall plots corresponding to the selected scenarios.

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